

Motion in two dimensions: vector representation and subtraction; relative velocity in a plane · CORE2B

Vector representation and subtraction: transfer

Intrinsic concept difficulty: MEDIUM

Question AUTHOR-VEC-03

In a common east/north frame, $P = (6, 0)$ m/s and $Q = (0, 8)$ m/s. Construct $P - Q$ graphically (reverse Q , then add tail-to-head) and state the resultant's components and magnitude.

Conditions:

1. Constant vectors; common east/north frame; free vectors may be translated without rotation.

[AUTHOR_CREATED · V3B relative-motion vertical slice; author-created candidate questions, not CBSE examination items \(V3B-Stress-Test-Prompt-Template.md Sec.3: BUILD REVIEW CORPUS\). Concept progression and arithmetic control adapted from V3B-Motion-Grade9-Example.md and the reviewed candidate microtopic library.](#)

[Hints and full answer](#)

Relative velocity in a plane: transfer

Intrinsic concept difficulty: HARD

Labelled extension, beyond the two-object baseline scope: a swimmer's velocity relative to the water and the water's velocity relative to the ground are both given. Finding the swimmer's velocity relative to the ground needs the same relation used throughout this unit, rearranged: if $v_{A/C} = v_{A/B} - v_{C/B}$ and the current is described as water relative to ground rather than ground relative to water, then $v_{\text{swimmer/ground}} = v_{\text{swimmer/water}} + v_{\text{water/ground}}$. This is addition, not the subtraction drilled above, and it involves three objects (swimmer, water, ground) instead of two -- decide which relation applies before calculating; do not assume every relative-motion question is a subtraction.

Question AUTHOR-REL-04

A swimmer's velocity relative to the water is $(0,3)$ m/s (straight across). The water's velocity relative to the ground (the current) is $(4,0)$ m/s. Find the swimmer's velocity relative to the ground, and name which relation you used to combine the two given velocities.

Conditions:

1. Constant velocities; common east/north frame; classical (non-relativistic) composition.
2. Labelled extension: three-object composition, beyond the two-object baseline scope.

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[Hints and full answer](#)

Question AUTHOR-REL-05

Using the M/N result above (M relative to N is $(-5,12)$ m/s), find the velocity of N relative to M without recomputing from the original velocities of M and N.

Conditions:

1. Same frame and instant as AUTHOR-REL-02.

[AUTHOR_CREATED · V3B relative-motion vertical slice; author-created candidate questions, not CBSE examination items \(V3B-Stress-Test-Prompt-Template.md Sec.3: BUILD REVIEW CORPUS\). Concept progression and arithmetic control adapted from V3B-Motion-Grade9-Example.md and the reviewed candidate microtopic library.](#)

[Hints and full answer](#)

Hints, answers and repair

Question AUTHOR-VEC-03

► Hint 1

► Hint 2

► Hint 3

$P-Q = (6,-8)$ m/s; magnitude 10 m/s.

10 m/s

1. Reverse $Q = (0,8)$ to get $-Q = (0,-8)$ m/s.
2. Translate $-Q$ to the head of P without rotating it, and add tail-to-head: $(6,0)+(0,-8) = (6,-8)$ m/s.
3. Cross-check against direct component subtraction: $(6-0, 0-8) = (6,-8)$ m/s -- both routes must agree.
4. Right-triangle bridge for the magnitude: $\sqrt{6^2+8^2} = 10$ m/s.

Check: The graphical and component routes produced the same $(6,-8)$ m/s.

Question AUTHOR-REL-04

► Hint 1

► Hint 2

$(4,3)$ m/s; magnitude 5 m/s.

5 m/s

1. Identify the relation: swimmer relative to ground = swimmer relative to water + water relative to ground (addition, not subtraction, because the current is already given as water-relative-to-ground).
2. Add the components: $(0,3) + (4,0) = (4,3)$ m/s.
3. Right-triangle bridge: magnitude = $\sqrt{4^2+3^2} = \sqrt{25} = 5$ m/s.

Check: If the current were zero, the result would reduce to the swimmer's own $(0,3)$ m/s, as it should.

Question AUTHOR-REL-05

► Hint 1

(5,-12) m/s; same magnitude (13 m/s) as M relative to N.

13 m/s

1. Apply the reversal relation directly to the already-known result: $v_{N/M} = -(v_{M/N}) = -(-5,12) = (5,-12)$ m/s.
2. No need to return to M and N's original velocities -- the reversal principle applies to any relative-velocity result.

Check: Adding $v_{M/N}$ and $v_{N/M}$ gives (0,0) m/s.